Lab 6, Act 2: An Interval for the Lethal Dose

How uncertain is our estimate of the LD50?

NRES 746

Answer key. Answers are in magenta.

The scenario

A pest manager has seen the fumigation results from Act 1 and wants to know the dose that kills half the beetles (the **LD50**) in mg/L, with a 95% interval. The model from Act 1 has no LD50 parameter; the LD50 is a function of a and b. In this Act you rewrite the model so that L, the LD50 on the log10 dose scale, appears as a parameter. You then build two intervals for L (one from the Hessian, one from the likelihood profile) and convert them to mg/L at the end ($\text{LD50} = 10^L$).

How uncertain is the LD50, and do the two methods agree?

Try Part 1 on paper first, before you open a laptop or start typing code.

Part 1: Plan it on paper (individual, ~20 min)

1a. Act 1 used the model

$$\log \frac{p}{1-p} = a + b\,x,$$

where p is the probability that a beetle dies at log10 dose x. Let L be the log10 dose at which $p = 0.5$. Find L in terms of a and b, and show that this model can be rewritten as $\log \frac{p}{1-p} = b\,(x - L)$. Then use Act 1's MLEs ($\hat{a} = -60.72$, $\hat{b} = 34.27$) to estimate L, and convert your estimate to mg/L.

$p = 0.5$ means log-odds $= 0$: $a + bL = 0$, so $L = -a/b$. Then $a + bx = b(x + a/b) = b(x - L)$.

$\hat{L} \approx 60.72/34.27 \approx 1.772$ (log10 dose) $\qquad$ $\text{LD50} = 10^{\hat{L}} \approx 59.1$ mg/L

1b. Write the negative log-likelihood, $\text{NLL}(L, b)$, for the eight doses. You can drop terms that don't depend on L or b.

$$\text{NLL}(L, b) = -\sum_{i=1}^{8} \left[k_i \log p_i + (n_i - k_i) \log(1 - p_i)\right], \qquad p_i = \frac{1}{1 + e^{-b(x_i - L)}}$$

(k_i killed of n_i exposed at dose i; the $\log \binom{n_i}{k_i}$ terms are dropped.)

1c. In plain English (not R code), describe step by step how you would compute the likelihood profile for L and read a 95% interval off it. (Recall how the profile differed from the slice in Act 1.)

Choose a grid of L values around $\hat{L}$. For each, hold L fixed and find the b that minimizes the NLL; record that minimum. Subtract the overall minimum NLL (at the MLE). The 95% interval runs from the smallest to the largest L whose value is 1.92 or less.

Checkpoint: finalize your plan (~5 min)

Look back over 1a-1c and fix anything that no longer looks right before you start coding.

Part 2: Fit and compare (laptop, ~50 min)

Each coding question is labeled with a recommendation:

- **Recommended: no AI:** try to build this yourself. R's help pages, web search, and classmates are fine.
- **AI optional/encouraged:** the result matters, but writing this code by hand isn't the point.

After each one, circle how you actually used AI. Using AI never costs points; only undisclosed use does.

What to turn in: your filled-in answers and your 2c figure as an image (upload in WebCampus). You don't need to submit your code.

The data are in `BeetleFumigation.csv` (columns `log10_dose`, `beetles`, `killed`), on the course website: https://kevintshoemaker.github.io/NRES-746/data/BeetleFumigation.csv. You can read it straight from that link:

```
data_url <- "https://kevintshoemaker.github.io/NRES-746/data/BeetleFumigation.csv"
beetle_data <- read.csv(data_url)
```

Or paste the data into R directly:

```
log10_dose <- c(1.6907, 1.7242, 1.7552, 1.7842, 1.8113, 1.8369, 1.8610, 1.8839)
beetles    <- c(59, 60, 62, 56, 63, 59, 62, 60)
killed     <- c(6, 13, 18, 28, 52, 53, 61, 60)
```

2a. *(Recommended: no AI.)* Write an NLL function whose first argument is `par = c(L, b)`. Check it: at $L = 1.75$, $b = 30$ it should return about 32.19 if you used `dbinom()`, or about -135.33 if you coded the formula yourself and dropped the constant. Then fit it with `optim(..., hessian = TRUE)`, starting from your values in 1a. Use `solve()` on the Hessian to get the variance-covariance matrix; the square roots of its diagonal are the standard errors. Report the 95% quadratic interval, $\hat{L} \pm 1.96\,\text{SE}$.

$\hat{L} =$ 1.7717 $\quad \hat{b} =$ 34.27 NLL at the MLE: 18.72 (or -148.81 without the constant)

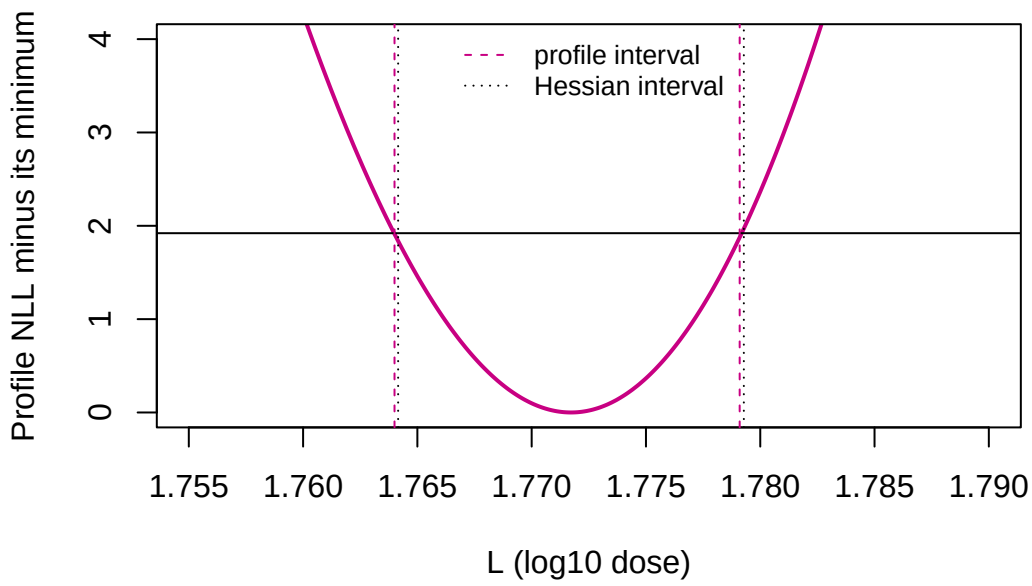
$\text{SE}(\hat{L}) =$ 0.0039 interval for L: 1.7642 to 1.7793

AI use (circle one): no AI / AI helped (e.g., debugging) / AI wrote the code

2b. *(Recommended: no AI.)* Implement your profile from 1c and report the 95% profile interval for L. How does it compare with 2a?

Profile interval for L: 1.7640 to 1.7791

Nearly identical to the quadratic interval from 2a.



AI use (circle one): no AI / AI helped (e.g., debugging) / AI wrote the code

2c. *(AI optional/encouraged.)* Plot your profile. Put L on the x-axis. On the y-axis, put the profile NLL at each value of L minus the smallest profile NLL, so the lowest point of the curve is at zero. Add a horizontal line at 1.92 and mark both intervals (from 2a and 2b). Then convert $\hat{L}$ and the profile interval to mg/L.

LD50 $= 10^{\hat{L}}$: 59.1 mg/L, from 58.1 to 60.1 mg/L

AI use (circle one): no AI / AI helped (e.g., debugging) / AI wrote the code

2d. Where did you use AI, if anywhere, and what did you have to check or fix? If you didn't use it, what was hardest to get working?

Open-ended.

If you finish early. *(AI optional/encouraged.)* The dose that kills 99% of beetles, L_{99}, satisfies $\log \frac{p}{1-p} = b\,(x - L_{99}) + \log \frac{0.99}{0.01}$. Profile L_{99} the same way and compare the width of its interval with the LD50's.

$\hat{L}_{99} \approx 1.906$ (80.5 mg/L); profile interval 76.8 to 85.3 mg/L, about 3 times as wide (log scale) as the LD50's.

AI use (circle one): no AI / AI helped (e.g., debugging) / AI wrote the code

Reflection (~10 min)

Suppose the pest manager wants a dose that kills 99% of beetles instead of 50%. Without computing anything, would you expect its interval to be wider or narrower than the LD50's? (Look at where the data sit in Act 1's figure.) The Hessian and profile intervals nearly agreed here. When would you expect them to disagree? Think back to the chytrid curve from Monday's WOD.

Also record any changes you made, new insights, and any "muddiest points" or topics you want to review.

Wider: the LD99 lies above the highest dose tested, where batches can only show "nearly all died," and it depends heavily on the slope. The Hessian and profile intervals disagree when the NLL isn't close to a parabola: small samples, estimates near a boundary (like prevalence near 0 in the chytrid WOD), or quantities outside the range of the data.